Modelling Organic Molecule Survival on Mars: Finding the Goldilocks Preservation Depth

A Computational Astrobiology Study Using Real Rover Data
Curiosity (Gale Crater) & Perseverance (Jezero Crater)

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Abstract

This study models the survival of glycine on the Martian surface under combined UV photolysis, ionizing radiation, and thermal degradation. A depth-dependent model identifies the subsurface depth maximising glycine half-life — the Goldilocks preservation depth. Theoretical parameters yield a depth of 0.81 cm. Real UV data from 769,179 validated Curiosity REMS measurements shifts this to 1.01 cm. Sol-by-sol analysis reveals the Goldilocks zone fluctuates between 0.81 and 1.11 cm with dust-modulated UV variation. Comparison with Perseverance MEDA-RDS data from Jezero Crater (2,550,808 measurements, 53 sols), after normalisation for inter-instrument differences, yields 1.01 cm vs 1.04 cm at Gale. A refined model with realistic constant subsurface temperature identifies 0.71 cm as the minimum effective UV shielding depth. Sensitivity analysis reveals activation energy and regolith UV attenuation length as the dominant model uncertainties.

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1. Introduction

Glycine (NH_2CH_2COOH), the simplest amino acid, is a key biosignature target — detected in meteorites and comets, suggesting the building blocks of life are widespread in the solar system. On Mars, the surface is exposed to intense UV radiation (no ozone layer), cosmic rays and solar energetic particles, and diurnal temperature swings of up to 120°C. Laboratory experiments have demonstrated that organic molecules degrade rapidly under Mars-like UV conditions (Stalport et al., 2009; Noblet et al., 2012).

Subsurface burial offers protection: UV flux attenuates exponentially with depth. However, the Martian geothermal gradient means thermal degradation increases with depth. An optimal depth exists where UV shielding is maximised but thermal degradation has not yet become dominant — the Goldilocks preservation depth. This project quantifies it at Gale Crater (Curiosity) and Jezero Crater (Perseverance) using real rover data. This mirrors two of MEDA's core science objectives: quantifying UV destructive potential and estimating subsurface

habitability (Rodriguez-Manfredi et al., 2021).

2. Theoretical Framework

2.1 First-order decay kinetics

Glycine degradation follows first-order kinetics — each mechanism acts on a fixed fraction of remaining molecules per unit time:

$$dC/dt = -k_{total}\, C \quad \rightarrow \quad C(t) = C_0\, e^{-kt} \qquad t_{1/2} = \ln(2) / k_{total}$$

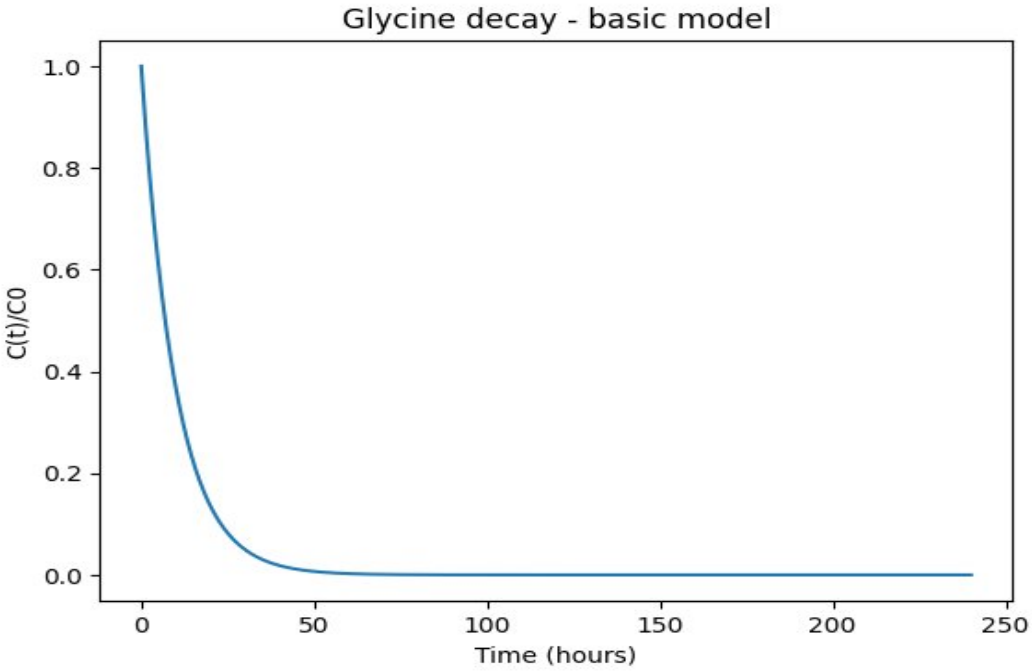

Figure 1. Exponential decay of normalised glycine concentration (placeholder k = 0.1/hr).

2.2 UV photolysis

$k_{UV} = \sigma \times F_{UV}$, where $\sigma = 1.2 \times 10^{-17}$ cm^2 is the absorption cross-section and F_{UV} is photon flux. UV attenuates with depth following Beer-Lambert law: $F_{UV}(z) = F_{surface}\, e^{-z/\lambda}$, where λ = 0.1 cm for Martian regolith, consistent with findings that UV is absorbed within the first millimetre of exposed rock (Cockell and Raven, 2004).

2.3 Ionizing radiation

$k_{rad} = \alpha \times D_{rate}$, where $\alpha = 2.5 \times 10^{-3}$ Gy^{-1} and $D_{rate} = 0.076 \times 10^{-3}$ Gy/hour (Hassler et al., 2014). Radiation attenuates as $k_{rad}(z) = k_{rad}\, e^{-z/3.0}$, consistent with Pavlov et al. (2012) who found solar cosmic rays destroy organics in the first 2 cm.

2.4 Thermal degradation

$k_{thermal} = A\, e^{-E_a/RT}$, where $A = 10^8$ s^{-1} and E_a = 45,000 J/mol. These glycine parameters were adopted from the project specification; independent experimental citation was not available. Subsurface temperature is modelled as $T(z) = 210 + 0.1z$ K in the initial model, refined to constant 210 K in Section 4.7.

2.5 Combined model

$$k_{total}(z) = k_{UV}(z) + k_{rad}(z) + k_{thermal}(z)$$

The Goldilocks depth z* maximises half-life — balancing UV shielding against thermal degradation.

3. Data Sources

3.1 Curiosity REMS — Gale Crater

UV flux from REMS (Gomez-Elvira et al., 2012), NASA PDS corrected UV dataset. Sols 1–100: 2,242,660 measurements at 1 Hz. After removing nighttime readings (flagged –999 for solar zenith > 55°): 769,179 valid measurements. UV_ABC (combined broadband A+B+C bands, W/m^2) used as F_{UV}. Conversion to photons/cm^2/s via $E = hc/\lambda$ with λ = 300 nm (an approximation across the 200–400 nm band; a spectral-weighted mean would be more precise).

3.2 Perseverance MEDA-RDS — Jezero Crater

UV from MEDA RDS (Rodriguez-Manfredi et al., 2021), NASA PDS4 archive. 53 sols within the first 89 sols: 2,550,808 measurements. RDS_TOP_3 channel used as broadband UV proxy.

3.3 Instrument comparison limitation

REMS measures integrated broadband UV; MEDA RDS uses narrow-band photodiodes. Direct comparison showed Jezero at 1.26 W/m^2 vs Gale at 15.43 W/m^2 — a 12× difference physically inconsistent with similar latitudes. Jezero data was normalised to Gale's mean, preserving relative sol-to-sol variation. Additionally, sol numbers are not temporally equivalent: Curiosity landed August 2012, Perseverance February 2021. The site comparison is therefore statistical (mean depth across each mission's early sols) rather than a day-by-day temporal correlation.

4. Results

4.1 UV dominance at the surface

Mechanism	Rate constant	Relative magnitude
UV photolysis (k_UV)	12.096 per hour	1
Ionizing radiation (k_rad)	1.90 × 10■■ per hour	~63 million × smaller

UV photolysis is approximately 63 million times faster than ionizing radiation, confirming that even minimal regolith shielding dramatically improves glycine survival.

4.2 Theoretical Goldilocks depth — 0.81 cm

Using theoretical UV flux (2.8×10^{14} photons/cm^2/s), the Goldilocks depth is 0.81 cm (Figure 2).

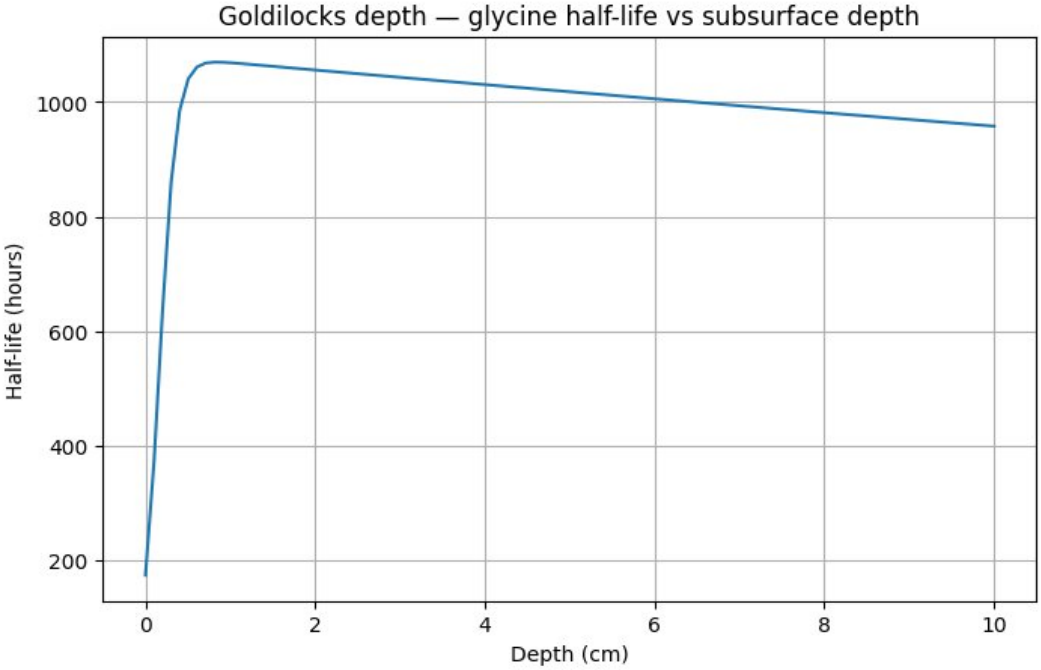

Figure 2. Theoretical Goldilocks depth: peak at 0.81 cm using assumed UV flux.

4.3 Real data correction — 1.01 cm

Curiosity REMS mean daytime UV: 15.43 W/m^2 = 2.33 × 10^{15} photons/cm^2/s — 731.8% higher than assumed. This shifts the Goldilocks depth to 1.01 cm (Figure 3). Note: the conversion assumes all broadband UV photons contribute to photolysis; in reality only photons exceeding glycine's bond dissociation energy (~3.5 eV) cause breakage, so k_{UV} may be overestimated.

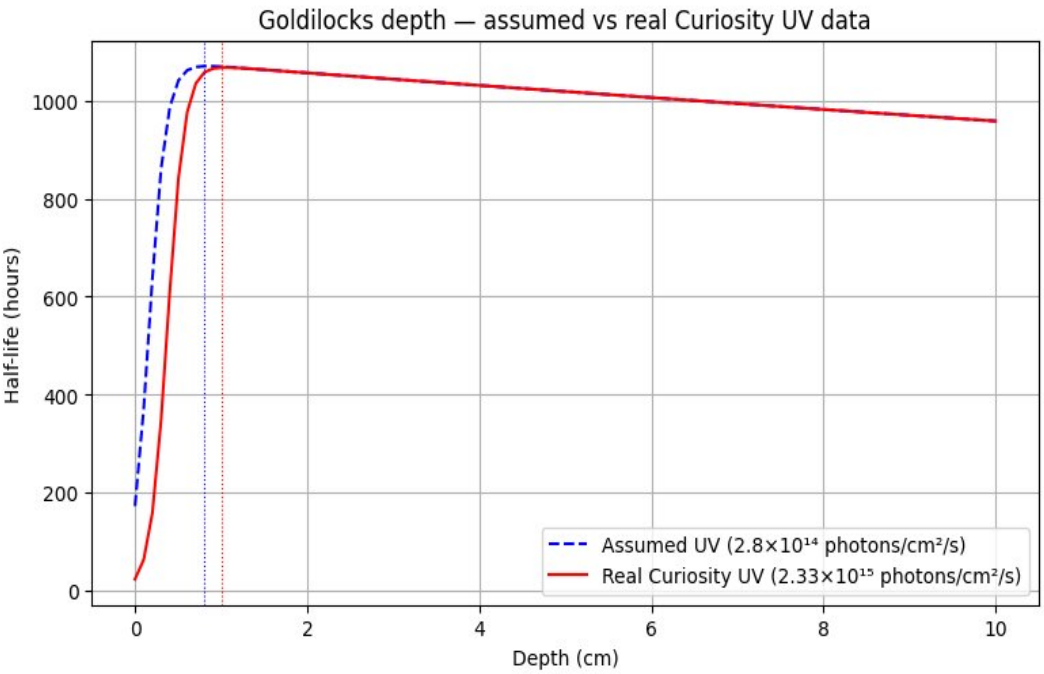

Figure 3. Theoretical (blue dashed, 0.81 cm) vs real Curiosity data (red solid, 1.01 cm).

4.4 Sol-by-sol variability

UV ranges from 2.01 W/m^2 (Sol 20, dust event) to 20.59 W/m^2 (Sol 89, clearest day) — a 10× variation driving Goldilocks depth fluctuation:

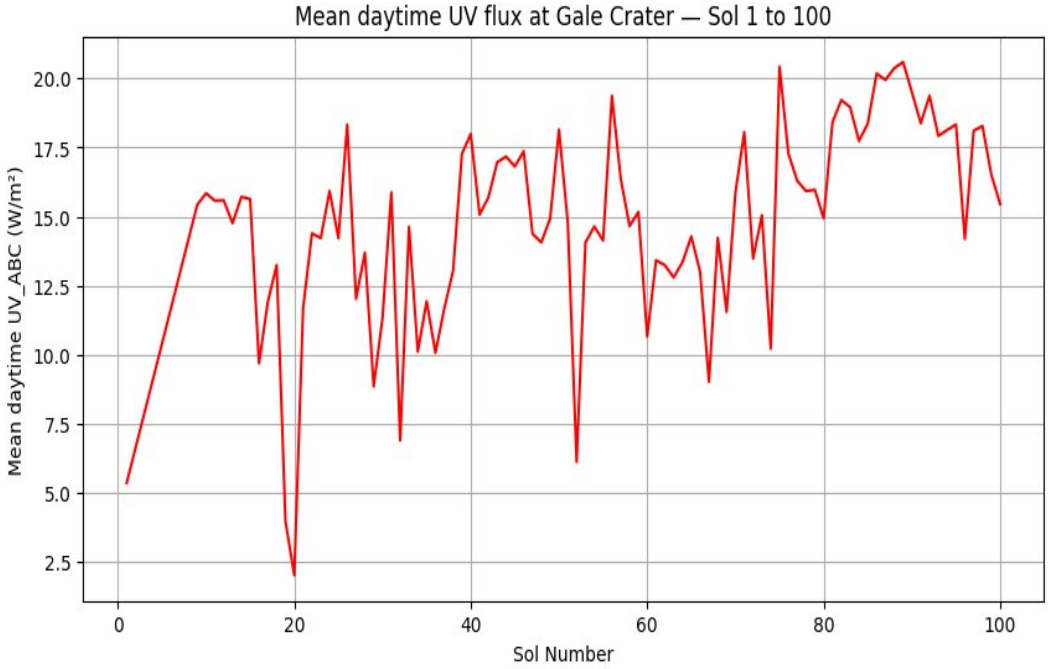

Figure 4. Mean daytime UV at Gale Crater, sols 1–100. Sol 20 dip indicates dust event.

Parameter	Value
Mean Goldilocks depth	1.04 cm
Minimum (dusty Sol 20)	0.81 cm
Maximum (clear Sol 89)	1.11 cm
Dynamic range	0.30 cm

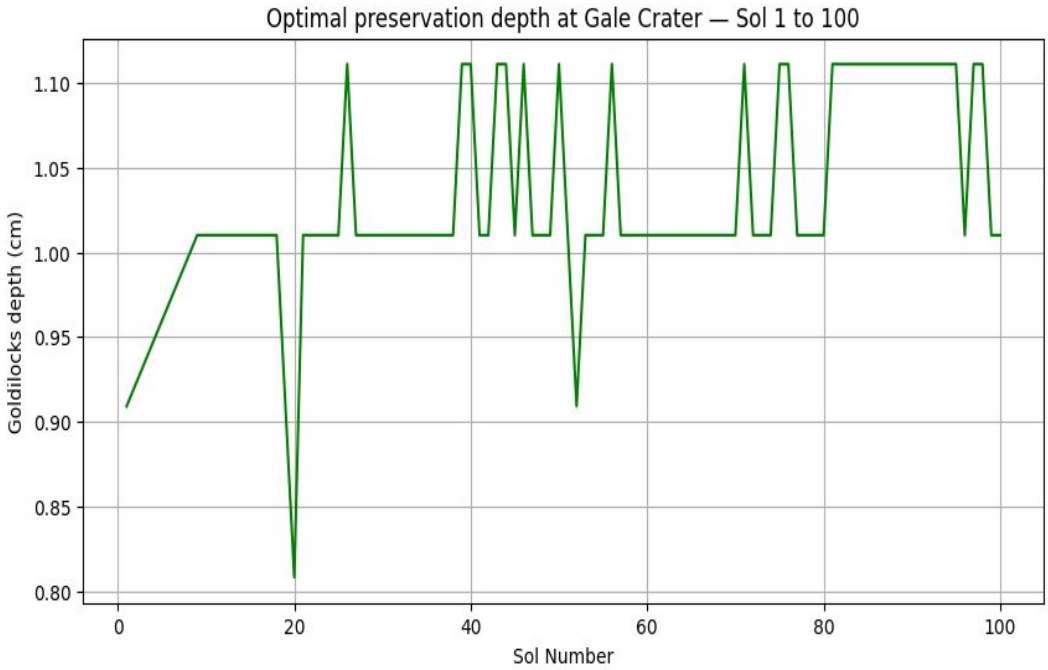

Figure 5. Sol-by-sol Goldilocks depth at Gale: 0.81–1.11 cm range.

4.5 Gale vs Jezero comparison

Normalised Jezero UV variability yields a mean optimal depth of 1.01 cm vs 1.04 cm at Gale. Sol numbers are not temporally equivalent between rovers (Curiosity Sol 1 = August 2012; Perseverance Sol 1 = February 2021), so this comparison is statistical rather than a day-by-day temporal correlation. The 0.03 cm difference reflects different dust climatologies and atmospheric conditions at the two sites (Figure 6).

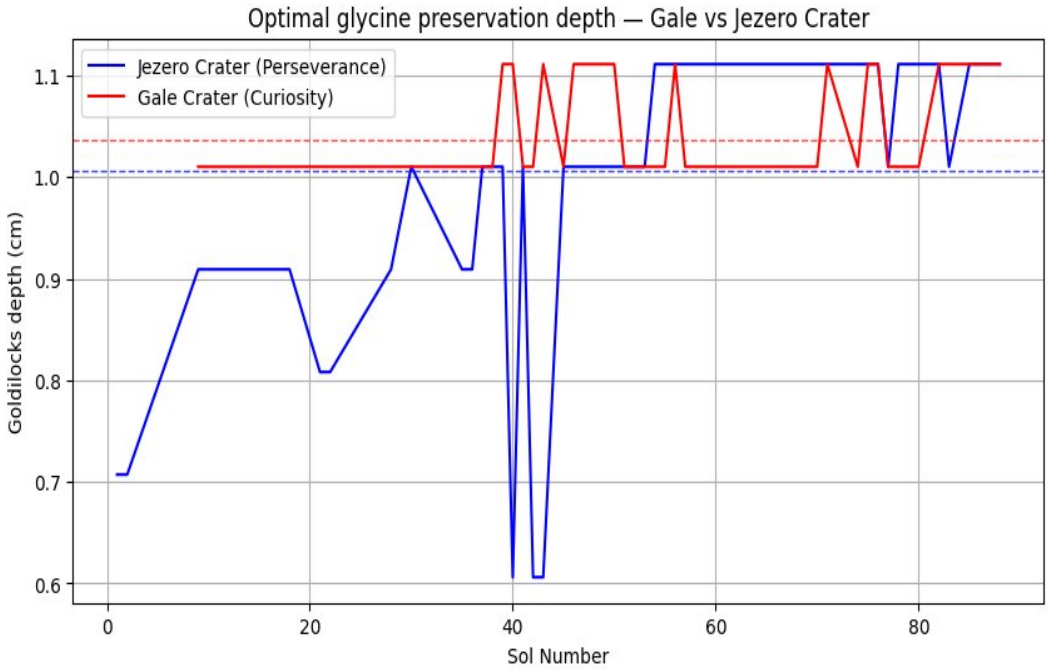

Figure 6. Goldilocks depth: Gale (red) vs Jezero (blue), plotted by mission sol number (not temporally equivalent). Dashed lines: means of 1.04 and 1.01 cm.

4.6 Sensitivity analysis

Each parameter was varied individually (0.5–2.0×, or 0.8–1.2× for E_a) while holding others constant:

Parameter	Variation	Depth range (cm)	Sensitivity
Ea (activation energy)	0.8–1.2×	0.61–1.62	1.01 cm
λ (UV attenuation)	0.5–2.0×	0.61–1.92	1.31 cm
σ (UV cross-section)	0.5–2.0×	1.01–1.11	0.10 cm
F_UV (UV flux)	0.5–2.0×	1.01–1.11	0.10 cm
α (rad sensitivity)	0.5–2.0×	1.01–1.01	0.00 cm

E_a and λ dominate uncertainty (shifting depth by >1 cm each), while α has zero effect, confirming radiation is irrelevant to preservation depth. The pre-exponential factor A was not independently varied; since A and E_a jointly govern the Arrhenius equation, A may contribute comparable uncertainty and should be included in future analyses.

4.7 Refined model — realistic subsurface temperature

The initial model used a geothermal gradient of 0.1 K/cm. The actual Martian geothermal gradient is ~5 K/km (0.005 K/m) — meaning temperature in the top 10 cm is effectively constant at 210 K. Correcting this yields a plateau rather than a peak: half-life rises sharply in the first centimetre then remains constant. The minimum effective shielding depth — where half-life reaches 95% of maximum — is 0.71 cm. Beyond this, glycine preservation is limited only by constant thermal degradation, with no penalty for going deeper in the top 10 cm.

5. Discussion

5.1 Implications for biosignature detection

The refined model identifies 0.71 cm as the minimum effective UV shielding depth. Combined with the sol-by-sol variability range of 0.81–1.11 cm from the initial model, a drilling target of approximately 1.0 cm represents a robust conservative choice across all observed weather conditions.

5.2 The dust paradox

Counterintuitively, dust storms may benefit organic preservation by reducing UV flux from 20.59 to 2.01 W/m^2, shifting the shielding threshold shallower. Extended dust storms create windows of enhanced near-surface preservation.

5.3 Site comparison

The consistency between Gale and Jezero (1.04 vs 1.01 cm) — despite different geology (154 km impact crater with 5 km central mountain vs ancient river delta) — suggests the preservation window may be broadly applicable across Mars.

6. Limitations

The initial model used an exaggerated geothermal gradient (0.1 K/cm vs realistic 0.005 K/m), corrected in Section 4.7. Only 100 Curiosity sols and 53 Perseverance sols were analysed. REMS and MEDA measure UV differently, requiring normalisation; sol numbers between missions are not temporally equivalent (different years, seasons), making the comparison statistical rather than temporal.

The UV-to-photon conversion assumes all broadband UV photons contribute to photolysis; only photons exceeding glycine's bond dissociation energy (~3.5 eV) actually cause breakage, so k_{UV} may be overestimated. The mean wavelength of 300 nm is approximate across the 200–400 nm UV_ABC band. The parameters $\sigma = 1.2 \times 10^{-17}$ cm^2 and $A = 10^8$ s^{-1} were adopted from the project specification without independent experimental citation. The sensitivity analysis varied E_a but not A; both jointly govern the Arrhenius equation and A may contribute comparable uncertainty.

The model treats regolith as homogeneous, ignoring rock layers, ice, or perchlorates. Temperature was modelled theoretically rather than from rover instruments. A minor unit

inconsistency exists: k_{UV} and $k_{thermal}$ are in per-second units while k_{rad} is per-hour; because k_{rad} is 63 million times smaller, this has negligible impact on depth results.

7. Conclusion

The optimal subsurface depth for glycine preservation on Mars is a dynamic quantity driven by UV flux variation. Using real Curiosity and Perseverance data, the Goldilocks depth is 1.01–1.04 cm at both sites, with sol-by-sol variability of 0.81–1.11 cm. UV photolysis dominates at 63 million times the rate of ionizing radiation. The refined model with realistic constant subsurface temperature identifies 0.71 cm as the minimum effective shielding depth.

We recommend a drilling target of approximately 1.0 cm as a conservative choice. The consistency between Gale and Jezero suggests broad applicability across Mars. Sensitivity analysis identifies activation energy E_a and regolith UV attenuation length λ as the highest-priority parameters for future experimental refinement.

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